Hinako Nishi

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EDUCATION

University of Michigan, Ann Arbor, Michigan
Bachelor's of Science in Engineering, *Mechanical Engineering*, *Minor in Mathematics* **May 2028**
GPA: 3.84/4.00, Dean's List x2
Coursework: Multivariable Calculus, Differential Equations, Linear Algebra, Introduction to Engineering (Underwater Vehicle Design), Solid Mechanics, Dynamics and Vibrations
Activities: UM Solar Car Team, ENGR100-600 Peer Mentor, Theta Tau Professional Engineering Fraternity

WORK EXPERIENCE

Japan Infra Waymark, *Mechanical Engineering Intern*, Japan **Jun 2026-Aug 2026**

- Designed and hand-built an amphibious inspection hovercraft across 6+ prototype iterations, reaching a 65 cm craft (down from 75 cm) that lifts a 3.5 kg payload at 50% motor output and clears 3 cm obstacles over 4 surface types (grass, gravel, water, and gentle slopes) for bridge infrastructure inspection.
- CAD-modeled and FDM 3D-printed 10+ core components in SolidWorks (lift-fan duct, ducted covers, 4 thruster stands) and integrated a flight controller, GPS, antennas, and 5 ESCs into a custom printed enclosure driving 5 motors off a single shared 4S battery to cut weight by 30%.
- Ran root-cause analysis on lift failures, sizing the system from first-principles (≈164 Pa cushion pressure, ≈500 CFM airflow, ≈120 W lift power); identified a propeller-limited fan and re-specced from a 7 in to 5 in prop, cutting the craft to 50% throttle at hover and running the motor well within thermal limits.
- Engineered a sewn ripstop-nylon skirt (36 pleats, 16 one-way drains) and a 4-thruster omnidirectional layout, and authored an 18-line manufacturing BOM (~¥309K) to assess unit cost and reproducibility.

Centre for Transformative Garment Production, *Research Intern*, Hong Kong **Jul 2024-Aug 2024**

- Engineered 4 iterative prototypes of a 2-finger robotic gripper using SOLIDWORKS and 3D printing, achieving 40% size reduction and improving camera visibility for garment detection.
- Enhanced mechanical precision by replacing lead screws with ball screws, optimizing timing belt-driven systems, resulting in 30% smoother finger motion and improved axis alignment across versions.
- Conducted a comparative study of 6+ robotic end-effector designs from academic papers and industrial applications; synthesized key design trade-offs and presented findings to the lab.

PROJECT EXPERIENCE

Underwater ROV Design Project, *Testing Lead* **Aug 2025-Dec 2025**

- Originated the ROV's pentagonal-prism frame concept in Onshape, a packaging decision that wrapped tightly around the stacked electronics and battery canisters, cut wasted internal volume, and helped keep total build cost to $256, well below comparable ROVs.
- Led thruster force testing in the Gorguze Family Lab using a calibrated load cell and 3:1 lever arm, measuring forward and reverse thrust across all 4 thrusters, pairing them by matched output (4.38 N and 4.25 N for surge, 1.94 N each for heave) to maximize symmetry and minimize unwanted yaw and pitch.
- Ran hydrostatic stability validation and Marine Hydrodynamics Lab surge and heave trials, tuning the buoyancy-ballast system to achieve self-righting stability and applying dimensional scaling to project full-scale performance against the 2.17 m/s lake-current requirement.

FIRST Robotics Team 9421, *Drive Team Captain/Engineering Team Member* **Aug 2023-May 2025**

- Led a 15-member engineering team in designing and building a competition robot within a six-week build season; served as Drive Team Captain, coordinating match strategy and pit operations through regional competitions and the 2024 FIRST World Championship.

SKILLS

Skills: Solidworks, Siemens NX, Onshape, ANSYS FEA, MATLAB, Mill, Lathe, 3D Printing, C++
Languages: Japanese (native), English (native), Mandarin (elementary proficiency)